

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv("QVI_data.csv")

df.head()
```

```
Out[2]:
```

	LYLTY_CARD_NBR	DATE	STORE_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_QT
0	1000	2018-10-17	1	1	5	Natural Chip Compny SeaSalt175g	
1	1002	2018-09-16	1	2	58	Red Rock Deli Chikn&Garlic Aioli 150g	
2	1003	2019-03-07	1	3	52	Grain Waves Sour Cream&Chives 210G	
3	1003	2019-03-08	1	4	106	Natural ChipCo Hony Soy Chckn175g	
4	1004	2018-11-02	1	5	96	WW Original Stacked Chips 160g	

```
In [3]: df["DATE"] = pd.to_datetime(df["DATE"])

df["MONTH"] = df["DATE"].dt.to_period("M")
```

```
In [4]: monthly = df.groupby(["STORE_NBR", "MONTH"]).agg(
    total_sales=("TOT_SALES", "sum"),
    customers=("LYLTY_CARD_NBR", "nunique"),
    transactions=("TXN_ID", "nunique")
).reset_index()
```

```
In [5]: monthly["txn_per_customer"] = monthly["transactions"] / monthly["customers"]
```

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In [6]: trial_stores = [77,86,88]

trial_data = monthly[monthly["STORE_NBR"].isin(trial_stores)]
control_data = monthly[~monthly["STORE_NBR"].isin(trial_stores)]
```

```
In [7]: def find_control_store(trial_store):

    trial = monthly[monthly["STORE_NBR"]==trial_store]

    correlations = []
```

```
for store in monthly["STORE_NBR"].unique():

    if store in trial_stores:
        continue

    control = monthly[monthly["STORE_NBR"]==store]

    merged = pd.merge(
        trial,
        control,
        on="MONTH",
        suffixes=("_trial", "_control")
    )

    corr = merged["total_sales_trial"].corr(merged["total_sales_control"])

    correlations.append((store,corr))

correlations = pd.DataFrame(correlations,columns=["STORE","CORR"])

best_store = correlations.sort_values("CORR",ascending=False).iloc[0]

return best_store
```

```
In [8]: for store in trial_stores:

        print("Trial Store:",store)

        print(find_control_store(store))
```

Trial Store: 77

```

c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:3057:
RuntimeWarning: Degrees of freedom <= 0 for slice
    c = cov(x, y, rowvar, dtype=dtype)
c:\ProgramData\anaconda3\Lib\site-packages\numpy\lib\_function_base_impl.py:2914:
RuntimeWarning: divide by zero encountered in divide
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STORE      31.0
CORR       1.0
Name: 30, dtype: float64
Trial Store: 86

```

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STORE      31.0
CORR       1.0
Name: 30, dtype: float64
Trial Store: 88

```

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```

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STORE      206.0
CORR        1.0
Name: 202, dtype: float64

```

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```

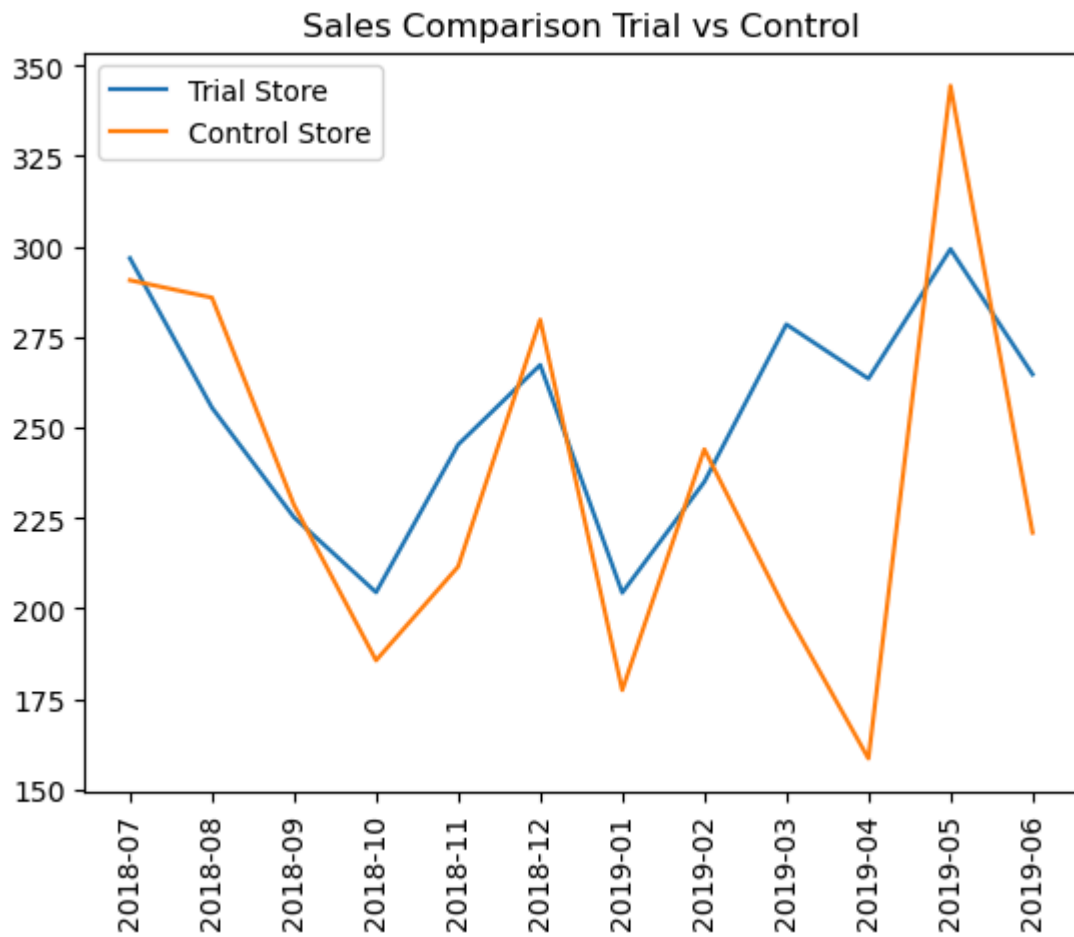
```

In [9]: trial = monthly[monthly["STORE_NBR"]==77]
        control = monthly[monthly["STORE_NBR"]==233]

plt.plot(trial["MONTH"].astype(str), trial["total_sales"], label="Trial Store")
plt.plot(control["MONTH"].astype(str), control["total_sales"], label="Control Store")

plt.xticks(rotation=90)
plt.legend()
plt.title("Sales Comparison Trial vs Control")
plt.show()

```



```
In [10]: comparison = pd.merge(
    trial,
    control,
    on="MONTH",
    suffixes=("_trial", "_control")
)

comparison["uplift"] = comparison["total_sales_trial"] - comparison["total_sales_control"]

comparison[["MONTH", "uplift"]]
```

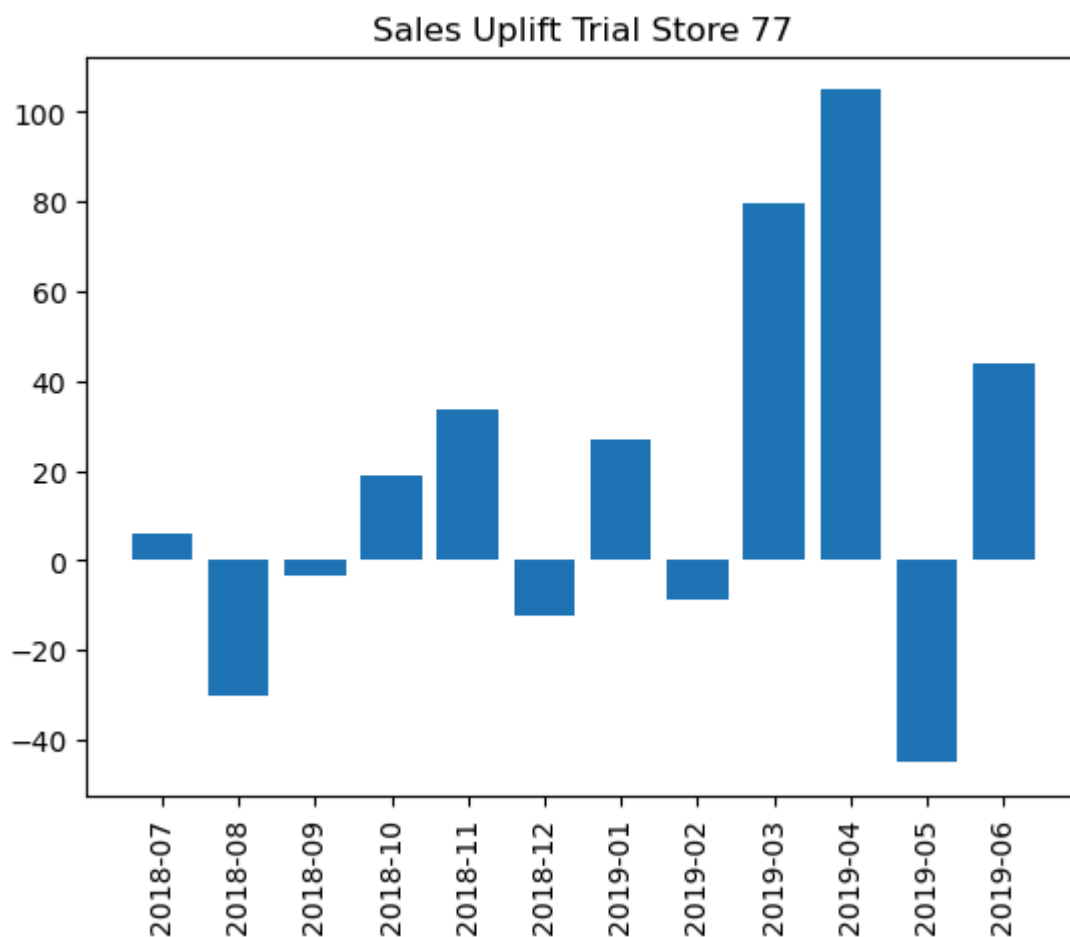
Out[10]:

	MONTH	uplift
0	2018-07	6.1
1	2018-08	-30.4
2	2018-09	-3.4
3	2018-10	18.8
4	2018-11	33.7
5	2018-12	-12.5
6	2019-01	26.9
7	2019-02	-9.0
8	2019-03	79.4
9	2019-04	104.9
10	2019-05	-45.1
11	2019-06	43.7

```
In [11]: plt.bar(comparison["MONTH"].astype(str),comparison["uplift"])
plt.xticks(rotation=90)

plt.title("Sales Uplift Trial Store 77")

plt.show()
```



In []:

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